

Experiment 1: Case Study

Problem Statement: Choose a real-world case study for KL ERP System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World case study an KL ERP System

1. Functional Requirements

Faculty

- Login securely
- View assigned courses
- Record student attendance
- Upload course materials
- Create and manage assignments
- Enter internal and examination marks
- View student academic performance
- Send announcements and notifications to students
- Manage class schedules
- Generate attendance and marks reports

Administrator

- Manage student records
- Manage faculty records
- Manage course information
- Create and update timetables
- Manage examination schedules
- Approve course registrations
- Manage fee and payment records
- Generate academic and administrative reports
- Manage user accounts and access permissions
- Monitor system activities and maintain data security

2. User Stories

User ID	User Story	Priority
US-01	As a faculty member, I want to record student attendance so that attendance records remain accurate.	High

User ID	User Story	Priority
US-02	As a faculty member, I want to upload course materials so that students can access learning resources.	High
US-03	As a faculty member, I want to create assignments so that student learning can be assessed.	Medium
US-04	As a faculty member, I want to enter marks so that students can view their academic performance.	High
US-05	As a faculty member, I want to send announcements so that students receive important academic updates.	Medium
US-06	As an administrator, I want to manage student records so that all information remains updated.	High
US-07	As an administrator, I want to manage faculty records so that staff information remains accurate.	High
US-08	As an administrator, I want to manage timetables so that academic schedules are organized efficiently.	High
US-09	As an administrator, I want to schedule examinations so that exams are conducted systematically.	Medium
US-10	As an administrator, I want to generate academic reports so that institutional performance can be monitored.	Medium
US-11	As an administrator, I want to manage user accounts and permissions so that the ERP system remains secure.	High
US-12	As an administrator, I want to monitor system security so that institutional data is protected.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Requirements may change frequently.
- Continuous feedback from students, faculty, and administrators.
- Incremental feature delivery.
- Faster development cycles.

- Easy bug fixing and maintenance.
- Suitable for large academic management systems.

Sprint Plan

Sprint Features

Sprint 1 Faculty Login & Attendance

Sprint 2 Course Materials

Sprint 3 Marks Module

Sprint 4 Student Records

Sprint 5 Reports

Sprint 6 Testing & Deployment

4. Programming Language Selection

Backend

- Java (Spring Boot)

Reasons

- Platform Independent
- Secure
- Scalable
- Enterprise-Level Application Development
- REST API Support
- Excellent Database Connectivity
- Strong Community Support

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Stripe/Razorpay Payment Gateway
- Git

- Docker
- AWS Cloud

5. System Architecture

Faculty

Administrator

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React.js Frontend

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REST API

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Spring Boot Backend

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MySQL Database

6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather customer requirements.
- Create Software Requirement Specification (SRS).
- Prepare use case diagrams.

Phase 2 – System Design

- Design database schema.
- Design UI wireframes.
- Create UML diagrams.
- Develop system architecture.

Phase 3 – Development

- Implement user authentication.
- Develop Faculty Module.
- Develop Administrator Module.
- Implement Attendance Module.
- Develop Examination Module.
- Develop Notification Module.

Phase 4 – Testing

Unit Testing

- Login

- Registration
- Attendance
- Marks Entry

Integration Testing

- Attendance + Results
- Fee Payment + Student Records

System Testing

- Complete ERP testing

User Acceptance Testing (UAT)

- Verify academic and administrative requirements.

Phase 5 – Deployment

- Deploy on Cloud Server.
- Configure MySQL Database.
- Deploy Spring Boot Application.
- Configure Web Server.
- Train Students, Faculty, and Administrators.

Phase 6 – Maintenance

- Fix defects.
- Improve performance.
- Add new features.
- Update security patches.
- Perform regular database backups.

7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement Verification
Performance	Load Testing
Security	Authentication & Penetration Testing
Reliability	Continuous Monitoring
Usability	User Feedback Survey
Scalability	Concurrent User Testing
Maintainability	Code Review & Documentation

8. Expected Outcomes

- Centralized academic management.
- Improved communication among students, faculty, and administrators.
- Online course registration and attendance management.

- Accurate examination and grading system.
- Secure online fee payment.
- Reduced paperwork and manual effort.
- Faster academic processes.
- Improved institutional efficiency.
- Secure and scalable ERP platform.
- Better decision-making through reports and analytics.